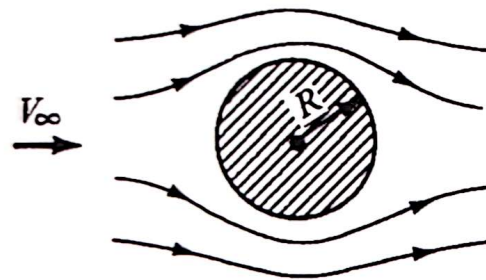


LIFTING FLOW OVER A CYLINDER

Review of
Spinning circular cyl.

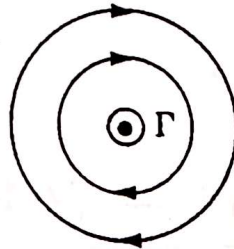
cylinder spinning about its axis

spinning



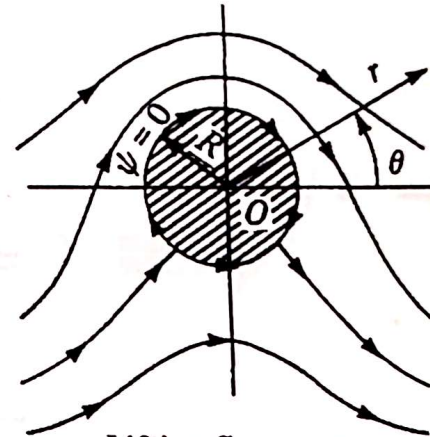
Nonlifting flow
over a cylinder

+



Vortex of
strength Γ

=



Lifting flow over
a cylinder

$$\psi_1 = (V_\infty r \sin \theta) \left(1 - \frac{R^2}{r^2} \right) \quad \psi_2 = \frac{\Gamma}{2\pi} \ln r + \text{const} \quad \text{Const} = -\frac{\Gamma}{2\pi} \ln R$$

$$\psi_2 = \frac{\Gamma}{2\pi} \ln \frac{r}{R}$$

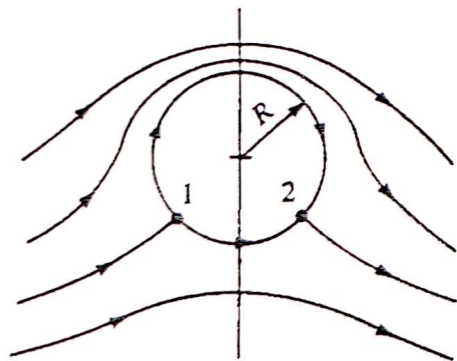
$$\psi = (V_\infty r \sin \theta) \left(1 - \frac{R^2}{r^2} \right) + \frac{\Gamma}{2\pi} \ln \frac{r}{R}$$

LIFTING FLOW OVER A CYLINDER

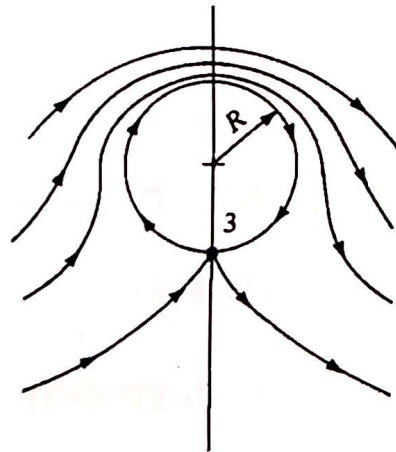
$$V_r = \left(1 - \frac{R^2}{r^2}\right) V_\infty \cos \theta \longrightarrow V_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta}$$

$$V_\theta = -\left(1 + \frac{R^2}{r^2}\right) V_\infty \sin \theta - \frac{\Gamma}{2\pi r} \quad V_\theta = -\frac{\partial \psi}{\partial r}$$

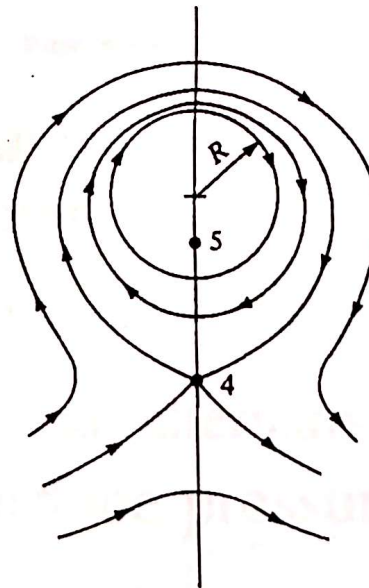
Stagnation points in the flow, set $V_r = V_\theta = 0 \longrightarrow \begin{cases} r = R \\ \theta = \arcsin\left(-\frac{\Gamma}{4\pi V_\infty R}\right) \end{cases}$



(a) $\Gamma < 4\pi V_\infty R$



(b) $\Gamma = 4\pi V_\infty R$



(c) $\Gamma > 4\pi V_\infty R$

Put these r & θ
in ψ Eqⁿ, you
Read relevant
discussions
get ψ_{body}

Figure 3.33 Stagnation points for the lifting flow over a circular cylinder.

The lift on the cylinder

$$C_L = \frac{L}{\frac{1}{2} \rho_\infty V_\infty^2 C}$$

$$C_p = \frac{p - p_\infty}{\frac{1}{2} \rho_\infty V_\infty^2}$$

$$C_l = C_n = \frac{1}{c} \int_0^c C_{p,l} dx - \frac{1}{c} \int_0^c C_{p,u} dx \quad x = R \cos \theta \quad dx = -R \sin \theta d\theta$$

$$C_l = -\frac{1}{2} \int_\pi^{2\pi} C_{p,l} \sin \theta d\theta + \frac{1}{2} \int_\pi^0 C_{p,u} \sin \theta d\theta$$

$C_{p,l}$ and $C_{p,u}$ are both given by the same analytic expression.

$$C_l = -\frac{1}{2} \int_0^{2\pi} C_p \sin \theta d\theta$$

$$\int_0^{2\pi} \sin \theta d\theta = 0$$

$$\int_0^{2\pi} \sin^3 \theta d\theta = 0$$

$$\int_0^{2\pi} \sin^2 \theta d\theta = \pi$$

$$C_l = \frac{\Gamma}{RV_\infty}$$

$$L' = \frac{1}{2} \rho_\infty V_\infty^2 2R \frac{\Gamma}{RV_\infty}$$

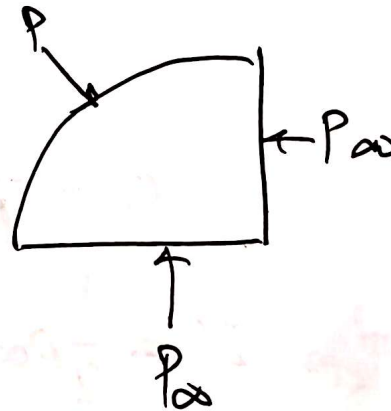
$$L' = \rho_\infty V_\infty \Gamma$$

Kutta-Joukowski theorem.

lift per unit span is directly proportional to circulation

$$D = \int_0^{\pi/2} (p \sin \theta) R \cos \theta d\theta - p_\infty R$$

$$y = R \sin \theta \rightarrow dy = R \cos \theta d\theta$$



LIFTING FLOW OVER A CYLINDER

The velocity on the surface of the cylinder $\left\{ \begin{array}{l} V_r = 0 \\ V = V_\theta = -2V_\infty \sin \theta - \frac{\Gamma}{2\pi R} \end{array} \right.$

$$C_p = 1 - \left(\frac{V}{V_\infty} \right)^2 = 1 - \left(-2 \sin \theta - \frac{\Gamma}{2\pi R V_\infty} \right)^2$$

$$C_p = 1 - \left[4 \sin^2 \theta + \frac{2\Gamma \sin \theta}{\pi R V_\infty} + \left(\frac{\Gamma}{2\pi R V_\infty} \right)^2 \right]$$

$$C_d = C_a = \frac{1}{c} \int_{LE}^{TE} (C_{p,u} - C_{p,l}) dy$$

$$y = R \sin \theta \quad dy = R \cos \theta d\theta$$

$$C_d = \frac{1}{c} \int_{LE}^{TE} C_{p,u} dy - \frac{1}{c} \int_{LE}^{TE} C_{p,l} dy$$

$$c = 2R$$

$$C_d = \frac{1}{2} \int_{\pi}^0 C_{p,u} \cos \theta d\theta - \frac{1}{2} \int_{\pi}^{2\pi} C_{p,l} \cos \theta d\theta$$

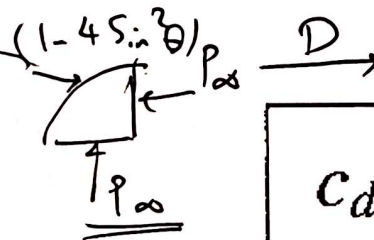
$$C_d = -\frac{1}{2} \int_0^{2\pi} C_p \cos \theta d\theta$$

$$\int_0^{2\pi} \cos \theta d\theta = 0$$

$$\int_0^{2\pi} \sin^2 \theta \cos \theta d\theta = 0$$

$$\int_0^{2\pi} \sin \theta \cos \theta d\theta = 0$$

Substituting Equation (3.126) into (3.130)



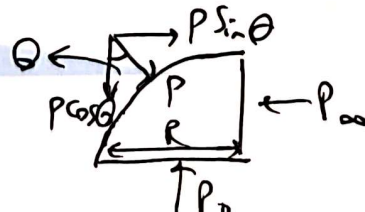
$$C_d = 0$$

for general $C_{p,u} \neq C_{p,l}$

use this

$$x = R \cos \theta \rightarrow dx = -R \sin \theta d\theta$$

$$L = p_\infty R - \int_{\pi}^{\pi/2} p \cos \theta (-R \sin \theta d\theta)$$



APPLIED AERODYNAMICS: THE FLOW OVER A CIRCULAR CYLINDER—THE REAL CASE

For very low values of Re , say, $0 < Re < 4$, the streamlines are almost (but not exactly) symmetrical, and the flow is attached, as sketched in Figure 3.45a. This regime of viscous flow is called *Stokes flow*; it is characterized by a near balance of pressure forces with friction forces acting on any given fluid element; the flow velocity is so low that inertia effects are very small. A photograph of this type

$Re = O(1)$, Only friction Drag

For $4 < Re < 40$, the flow becomes separated on the back of the cylinder, forming two distinct, stable vortices that remain in the position shown in Figure 3.45b. A photograph of this type of flow is given in Figure 3.47, where $Re = 26$.

As Re is increased above 40, the flow behind the cylinder becomes unstable; the vortices which were in a fixed position in Figure 3.45b now are alternately shed from the body in a regular fashion and flow downstream. This flow is sketched in

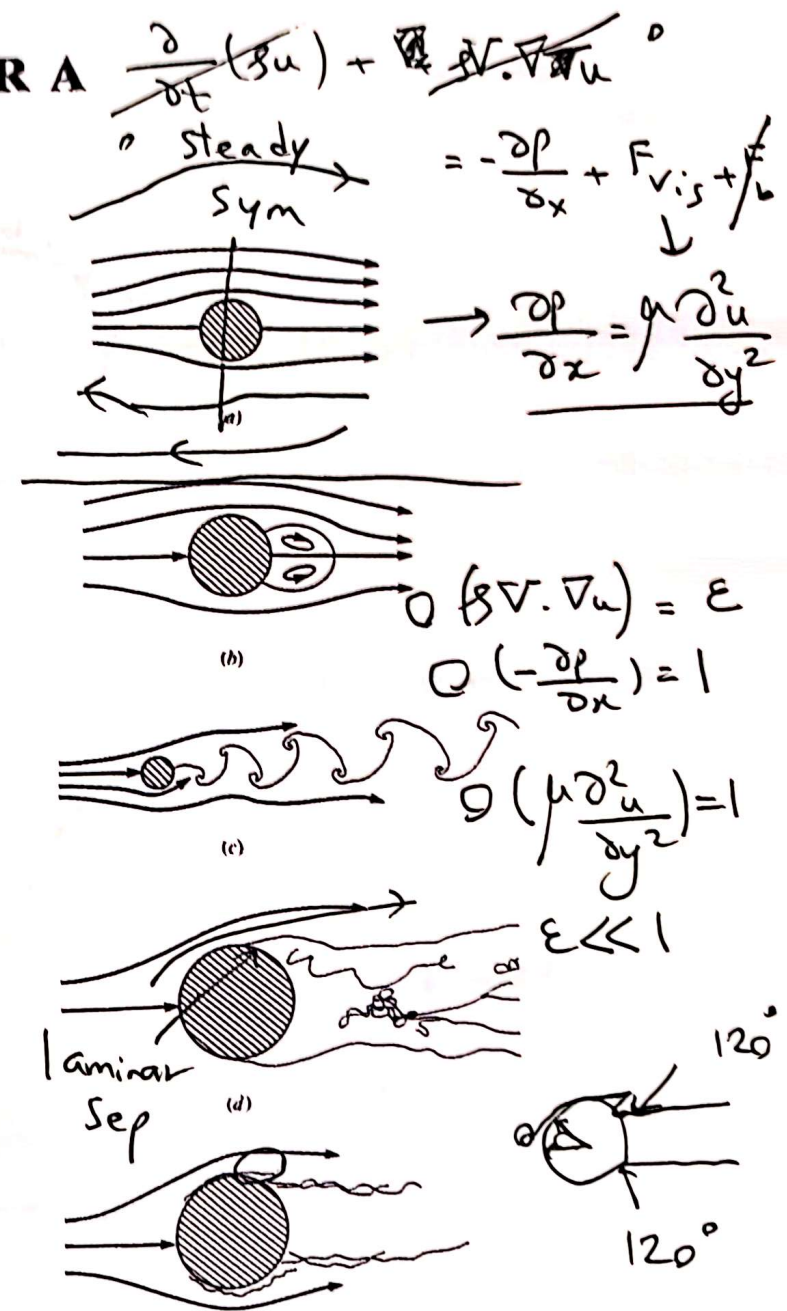
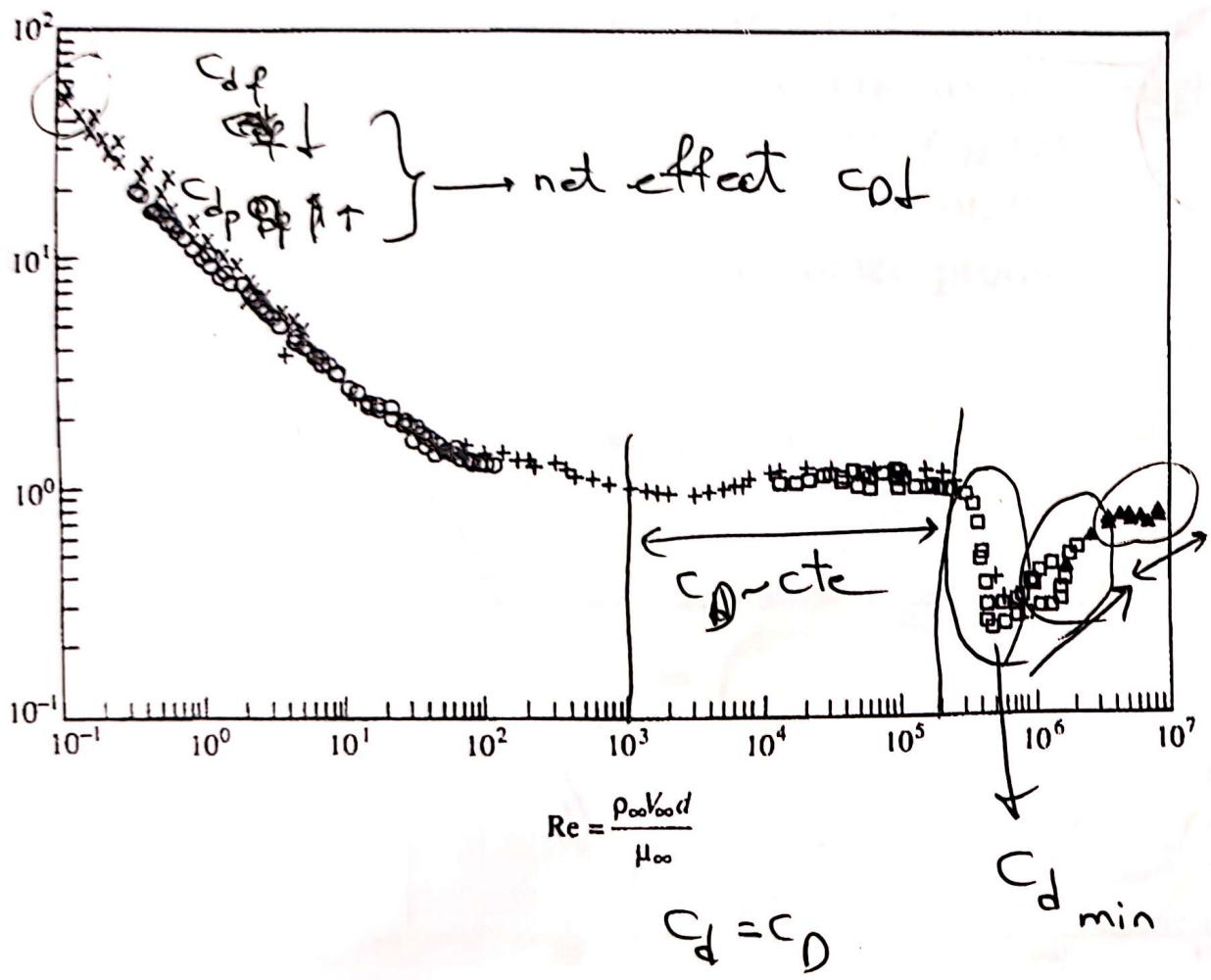
Karman vortex street,

APPLIED AERODYNAMICS: THE FLOW OVER A CIRCULAR CYLINDER—THE REAL CASE

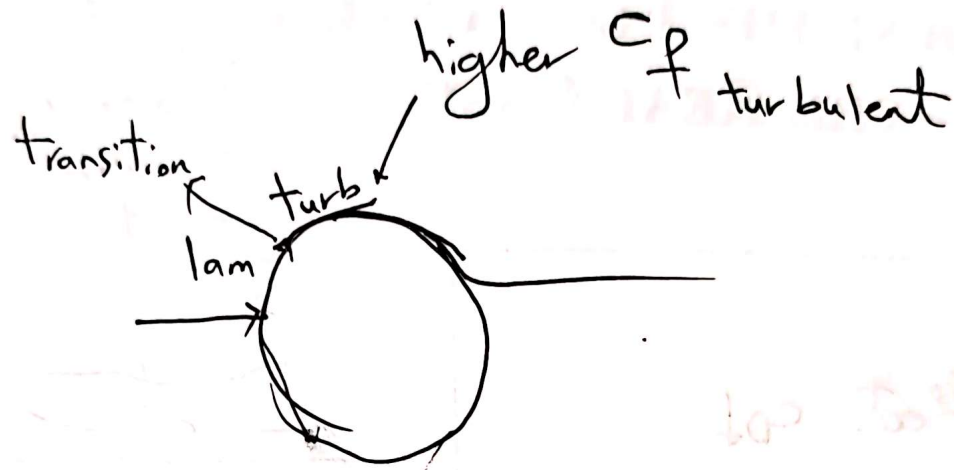
As the Reynolds number is increased to large numbers, the Karman vortex street becomes turbulent and begins to metamorphose into a distinct wake. The laminar boundary layer on the cylinder separates from the surface on the forward face, at a point about 80° from the stagnation point. This is sketched in Figure 3.45d. The value of the Reynolds number for this flow is on the order of 10^5 . Note, from Figure 3.44, that C_D is a relatively constant value near unity for $10^3 < Re < 3 \times 10^5$.

For $3 \times 10^5 < Re < 3 \times 10^6$, the separation of the laminar boundary layer still takes place on the forward face of the cylinder. However, in the free shear layer over the top of the separated region, transition to turbulent flow takes place. The flow then reattaches on the back face of the cylinder, but separates again at about 120° around the body measured from the stagnation point. This flow is sketched in Figure 3.45e. This transition to turbulent flow, and the corresponding thinner wake (comparing Figure 3.45e with Figure 3.45d), reduces the pressure drag on the cylinder and is responsible for the precipitous drop in C_D at $Re = 3 \times 10^5$

APPLIED AERODYNAMICS: THE FLOW OVER A CIRCULAR CYLINDER—THE REAL CASE



$D_{min} = C_{d \min} \frac{\rho}{2} V_{\infty}^2$
 not necessarily min



APPLIED AERODYNAMICS: THE FLOW OVER A CIRCULAR CYLINDER—THE REAL CASE

$$Re = \frac{\rho u D}{\mu} = \frac{1 \times 1 \times u}{10^{-5}} = 3 \times 10^6$$

For $Re \gtrsim 3 \times 10^6$, the boundary layer transits directly to turbulent flow at some point on the forward face, and the boundary layer remains totally attached over the surface until it separates at an angular location slightly less than 120° on the back surface. For this regime of flow, C_D actually increases slightly with increasing Re because the separation points on the back surface begin to move closer to the top and bottom of the cylinder, producing a fatter wake, and hence larger pressure drag.

$$u = 30 \text{ m/s}$$

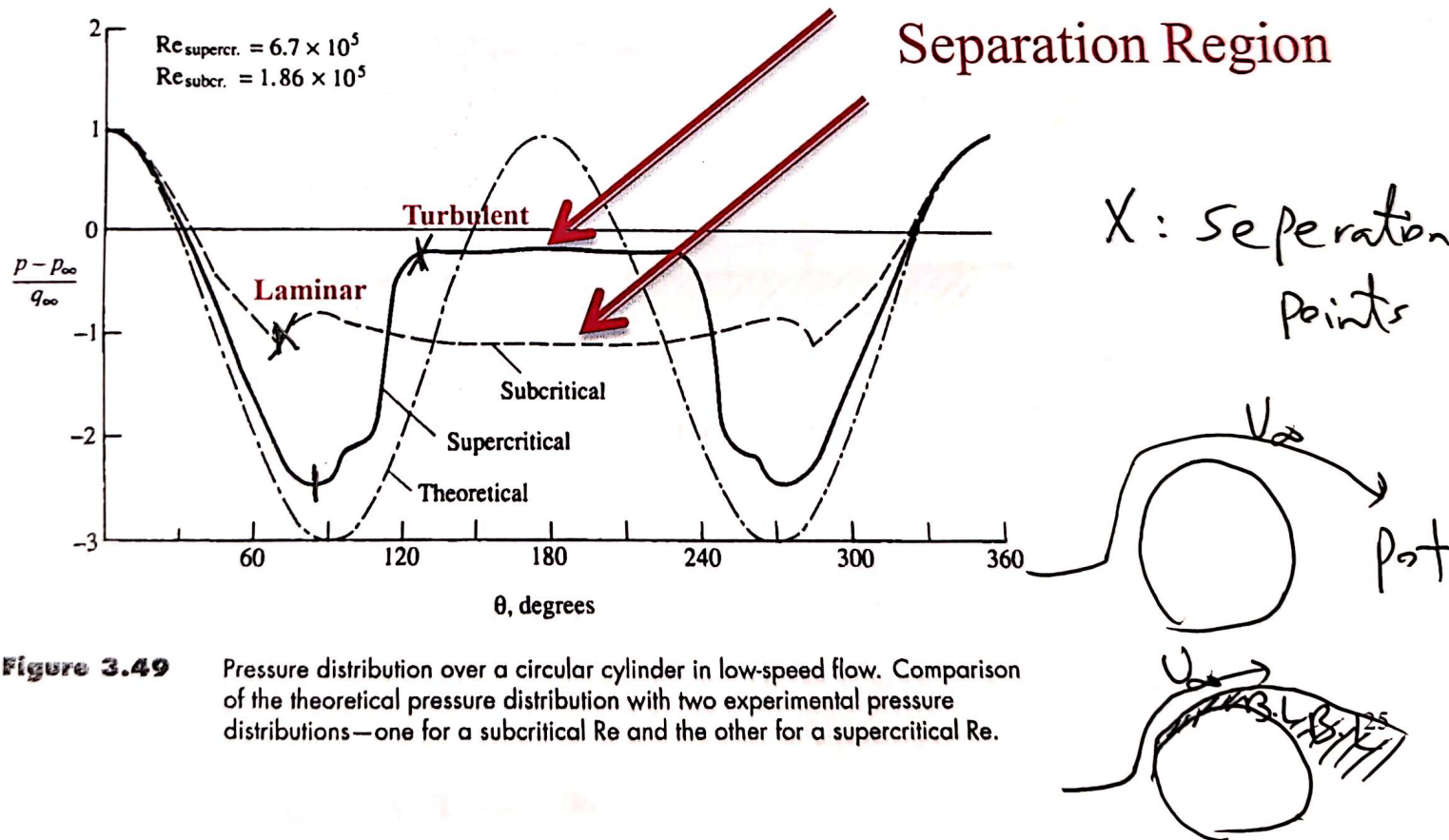
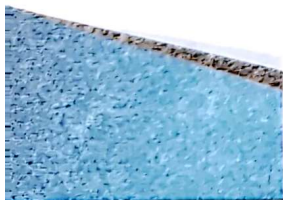


Figure 3.49 Pressure distribution over a circular cylinder in low-speed flow. Comparison of the theoretical pressure distribution with two experimental pressure distributions—one for a subcritical Re and the other for a supercritical Re .



Laminar & Turbulent BL

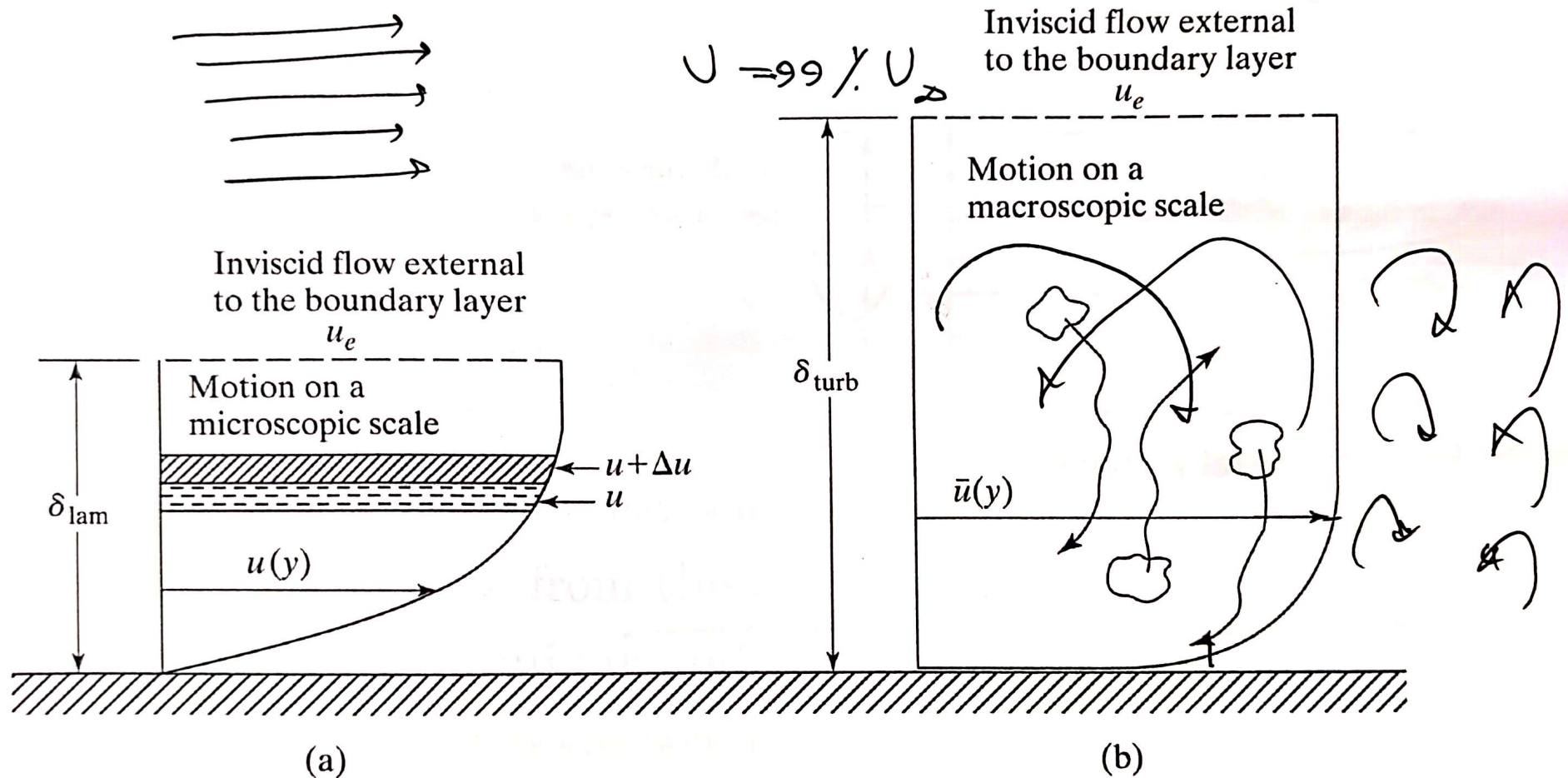
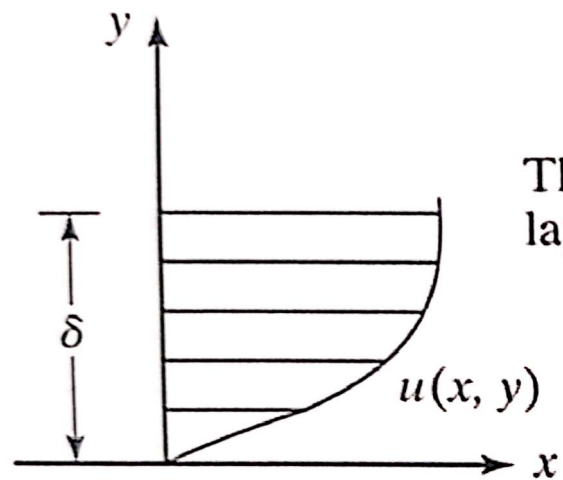


Figure 3.4: Laminar and turbulent boundary layer, Fig 4.2 from AE-BC

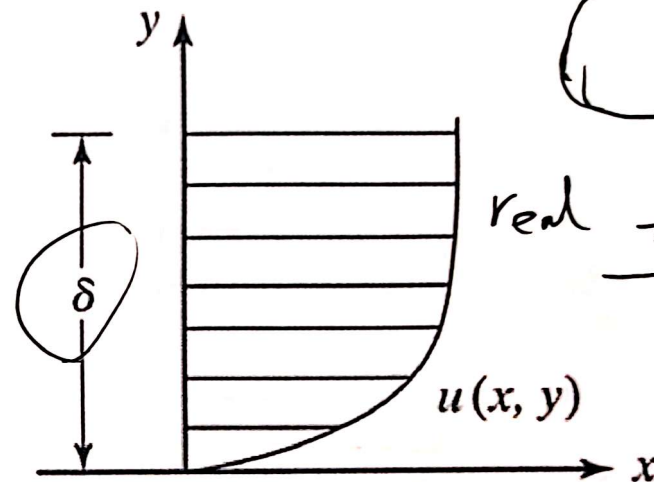
- ▶ Laminar → move in lamina or layers
- ▶ Turbulent → large-scale mixing - brings more momentum closer to the wall resulting in a fuller velocity profile

Laminar & Turbulent BL



Laminar boundary layer

The boundary layer thickness



Turbulent boundary layer

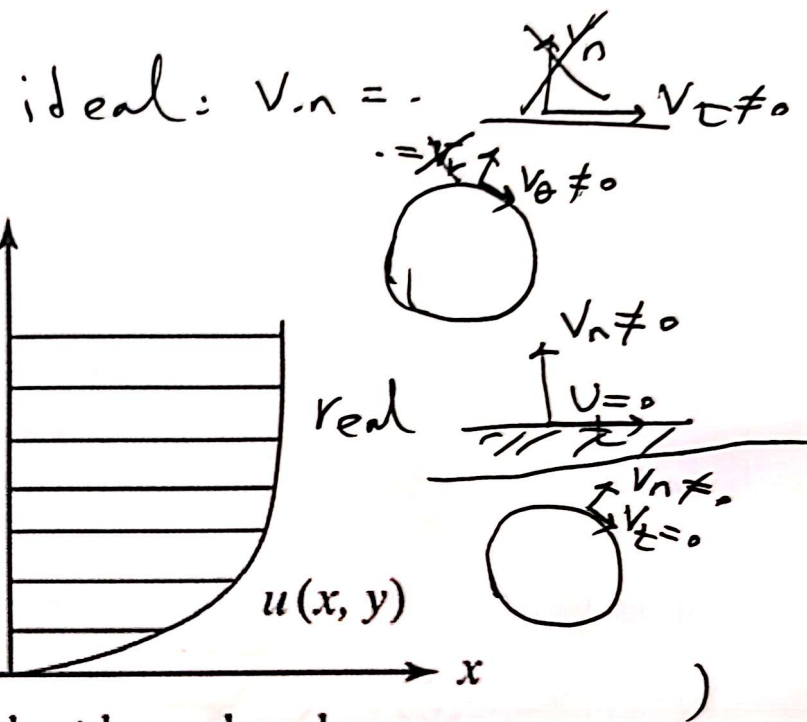


Figure 3.3: Laminar and turbulent boundary layer, Fig 2.14 from AE-BC

- ▶ Boundary layers arise from the no-slip boundary condition
- ▶ Two broad types - laminar and turbulent
- ▶ In general, turbulent boundary thickness $\delta_{turb} >$ laminar boundary thickness δ_{lam}
- ▶ Note δ also referred to as δ_{99} is the y location such that $u(y = \delta) = 0.99U_{\infty}$

Laminar & Turbulent BL

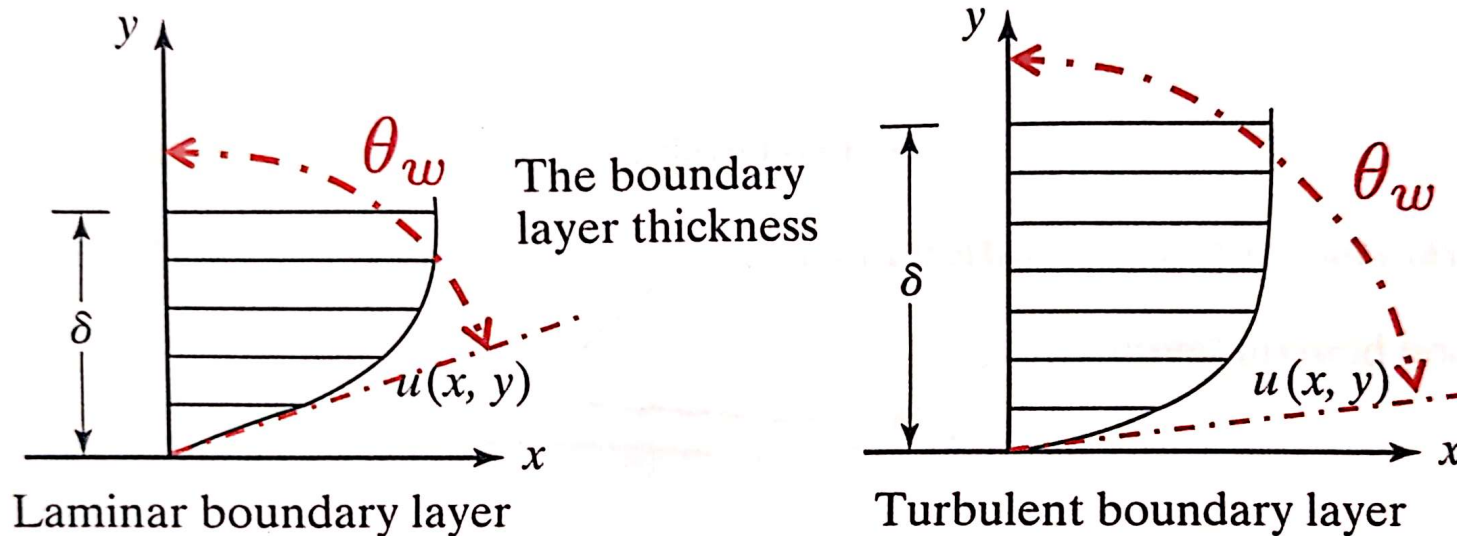


Figure 3.6: Wall shear stress, modified Fig 2.14 from AE-BC

Turbulent boundary layers have greater $[\partial u / \partial y]_{y=0}$ than laminar boundary layers. Hence, turbulent boundary layers have great wall shear stress and thus skin friction drag. Note,

$$C_{D\ sf} \propto C_f \frac{S_{wetted}}{S} \quad (3.2)$$

where S_{wetted} is the area “wetted” by the fluid and $C_{D\ sf}$ the coefficient of drag due to skin friction

Boundary layer over an airfoil

$$D = - \oint \underbrace{(\rho \mathbf{V} \cdot \mathbf{n})}_{\dot{m}} \underbrace{u}_{u} ds$$

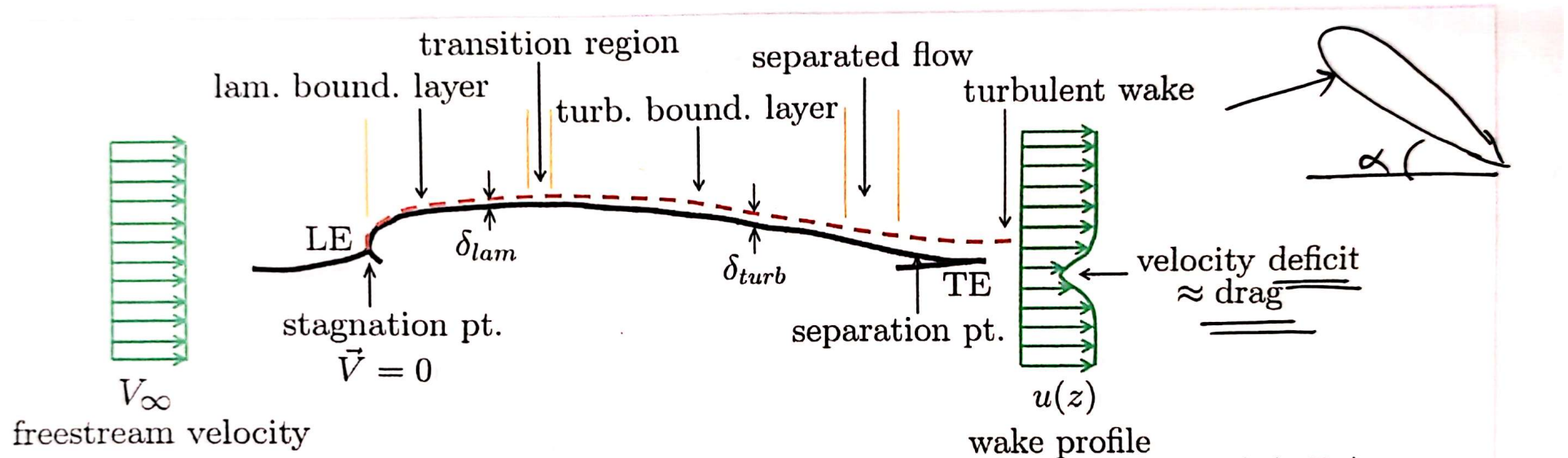


Figure 3.8: Boundary layers over airfoils partially adapted from AA-DA

- ▶ The boundary layer generally start of as laminar close to the LE at the stagnation point where $\vec{V} = 0$
- ▶ The stagnation pt. will vary with α
- ▶ Only the top surface is shown in Fig. 3.10, similar behavior is seen on the bottom surface

boundary layer over an airfoil

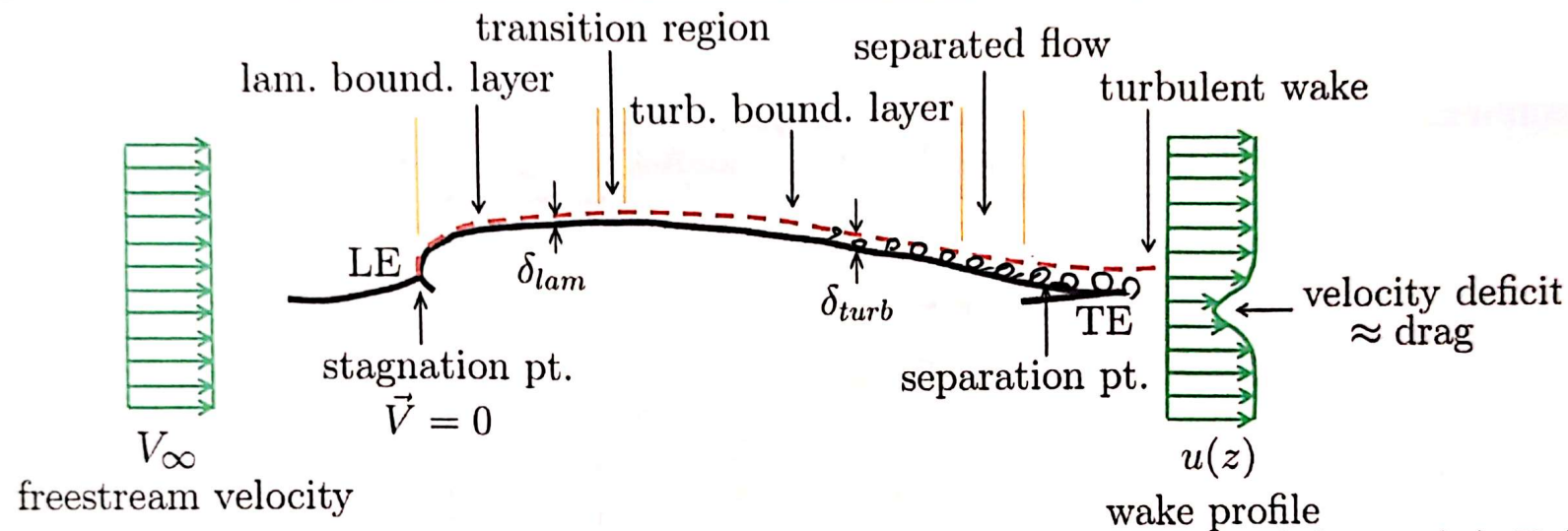


Figure 3.10: Boundary layers over airfoils partially adapted from AA-DA

- Due to no slip the boundary layer removes momentum from the freestream velocity which shows up as the velocity deficit in the wake. This velocity deficit is proportional to the drag.

low separation effects

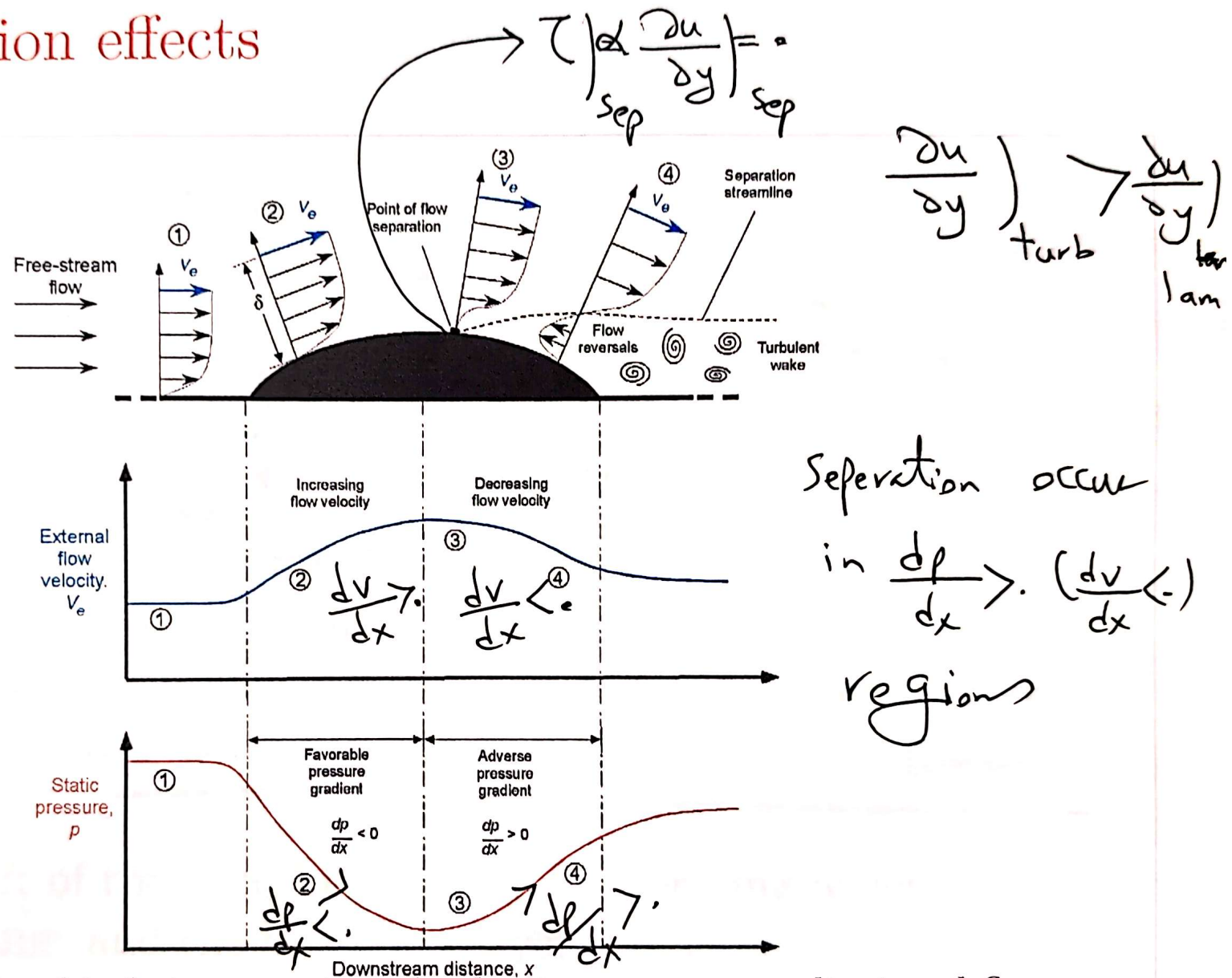


Figure 3.11: Relationship between external velocity, pressure gradient and flow separation. From IAFV-JGL

low separation over an airfoil

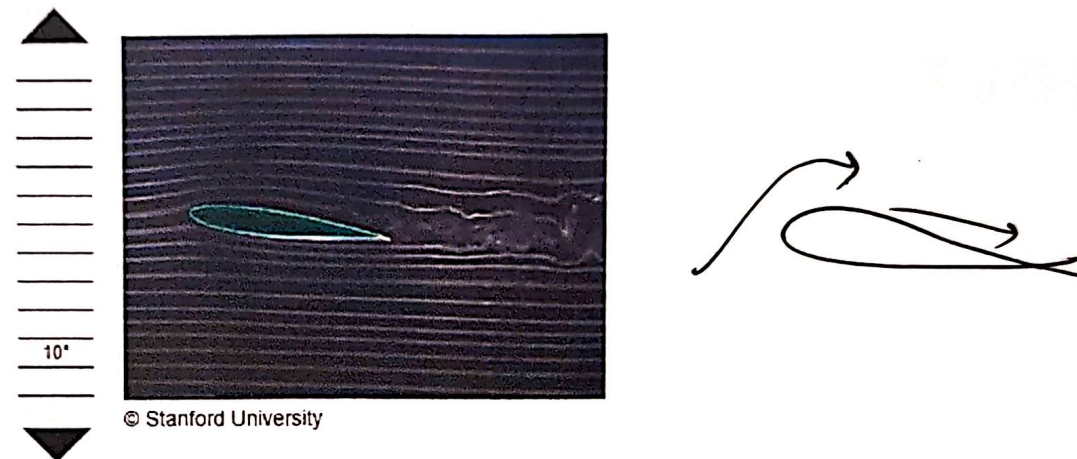


Figure 3.12: Flow separation over an airfoil, From MFM

- ▶ Over the leading edge of an airfoil the flow is accelerating which means U_e is increasing and the boundary layer profile becomes (turbo) “fuller” while p_e is decreasing – favorable pressure gradient
 $\partial p_e / \partial s < 0$.
- ▶ Over the aft of the airfoil the flow is decelerating which means U_e is decreasing and the boundary layer profile becomes less and less “full” while p_e is increasing – adverse pressure gradient
 $\partial p_e / \partial s > 0$.

